

Chapter 3 Critical Thinking Questions (page 70: #1 - #5 and #11):

1. a) employeeAge, test1, _value, total\$
b) 2ndPlace (cannot start with a number)
 user@id (@ is simply an illegal special character)
 my variable (has a space)
 class (a reserved keyword in Java)
 2. a) int numBeads;
 numBeads = 5;
b) int numBeads = 5;
 3. a) 13
b) 11
 4. a) int
b) float
c) int
d) double
e) boolean
f) char
 5. a) Primitive data types are the foundational building blocks of Java. They have single, simple values and have no built-in methods or complex behavior. Abstract data types are conceptual models. Ex: list is an ADT because it conceptually defines operations such as add(), remove(), and get().
b) A class is like a blueprint. It acts as a customized data type that dictates what fields (variables) and methods (functions) its instances will have. An object is physical, like a house built from the blueprint. It contains real values in its variables and can actively execute the methods defined by its class.
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11. a) $y = (\text{int}) (j * k); \quad = 8$
b) type casting is not necessary because j is an int and k is a double so java automatically promotes int to double before performing the math $= 8.0$
c) type casting is also not necessary because k is a double and the product is also a double meaning the data types match perfectly. $= 2.56$
d) $j = (\text{int}) \text{Math.round}(k); \quad = 2$
e) Type casting is not necessary because you are converting an int (less accurate) to a double (more accurate). A double can hold any value an int can hold without losing information. $= 5.0$
f) Type casting is also not necessary for this statement because both j and 3 are integers and their sum is also an int. Thus the data types match and data would not be lost. $= 8$